

NO GUESSWORK LIGHTING

No Guesswork Lighting

Documentation Suite

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Read the light. Know the setup.

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PART I

Quick Start Guide

Get your first lighting plan in under 5 minutes

Quick Start Guide

No Guesswork Lighting eliminates the trial-and-error of photographic setup. Tell the app what you want to shoot and receive a precise, actionable lighting plan — no prior experience required.

What You Need

- Any modern smartphone or browser (Chrome, Safari, Firefox)
- At least one light source — a window, lamp, or portable strobe
- Optional: reflector or diffusion panel for better results
- An NGW account — free tier available, no credit card required

Step 1 — Open the App

Navigate to the NGW app or open the installed PWA. On first visit you land on the Welcome screen. Tap **Try It Free** or **Get Started** to proceed.

✓ Tip

Signing in is not required to explore, but saving setups and accessing advanced features requires a free account.

Step 2 — Choose an Entry Point

The app home screen offers three ways to start:

1. **Analyse a photo** — Upload a reference image; NGW identifies the lighting pattern and tells you how to recreate it
2. **Browse Proven Setups** — Pick from a library of curated setups sorted by subject and mood
3. **Build from Scratch** — Use the Setup Wizard to describe your subject, space, and desired look

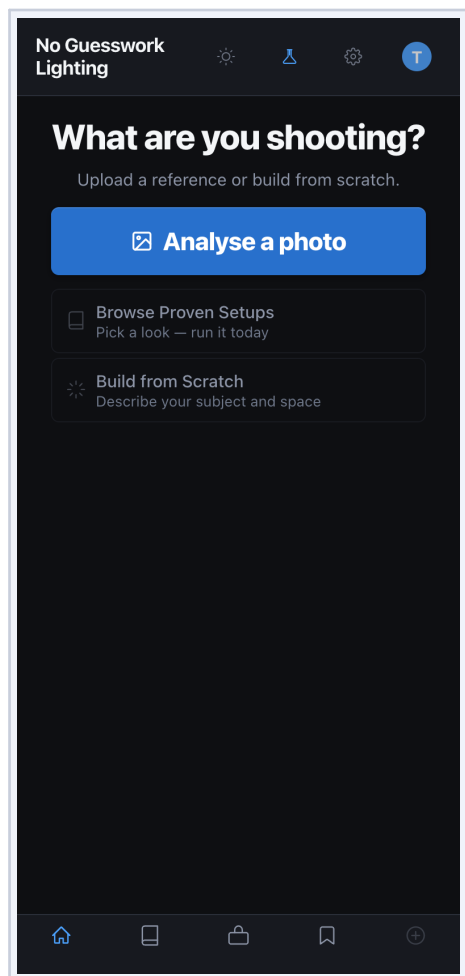


Figure 1 — App home screen: three entry points for starting a lighting plan

Step 3 — Review Your Lighting Plan

NGW returns a complete lighting plan. The Results screen shows:

- **Best Match** — Recommended system with confidence score
- **Setup** — Per-light: modifier, position, distance, power
- **Camera Settings** — Aperture, ISO, shutter, white balance
- **Diagram** — Top-down floor plan drawn to scale
- **Quick Fixes** — On-set troubleshooting for common problems

Step 4 — Read the Diagram

Tap the **Diagram** tab on the Results screen for a top-down floor plan showing exact light placement relative to your subject. Tap any light icon to highlight its setup card.

✓ Tip

If you've entered your room dimensions in the Room Planner, all distances in the diagram are shown in real-world feet or meters.

Step 5 — Save or Share

Tap the share icon to copy the setup as formatted text, send via SMS, or email to a collaborator. Signed-in users can save the setup to *My Setups* for instant recall on any device.

Bonus: Reference Photo Analysis

Have a lighting reference you want to replicate? Tap **Analyse a photo** on the home screen, upload the image, and NGW's 35-pass vision pipeline identifies the pattern and generates a recreation guide.

- Supported: JPEG, PNG, WebP, HEIC, TIFF (max 10 MB)
- Best results: well-exposed portrait with visible shadows and catchlights
- Returns: pattern name, modifier type, placement, fill strategy, step-by-step guide

PART II

Operations Manual

Complete guide to every screen and workflow

Operations Manual

This section documents every screen in the NGW application — controls, data flow between screens, and recommended workflows for common shooting scenarios.

Navigation

Navigation is managed via the bottom bar (five icons) and contextual action buttons within each screen.

Tab	Icon	Purpose
Home	House	Start a new setup, analyse a photo, or open a recent setup
My Kit	Camera	Manage your gear inventory
Kit Bag	Briefcase	Browse and run saved setups
Saved	Bookmark	Access reference images and saved sessions
More	Plus	Access Room Planner, Settings, Recipes, and Lab (admin)

Home Screen

The home screen is the app's primary hub. It presents three entry points and a quick-access row of recent setups.

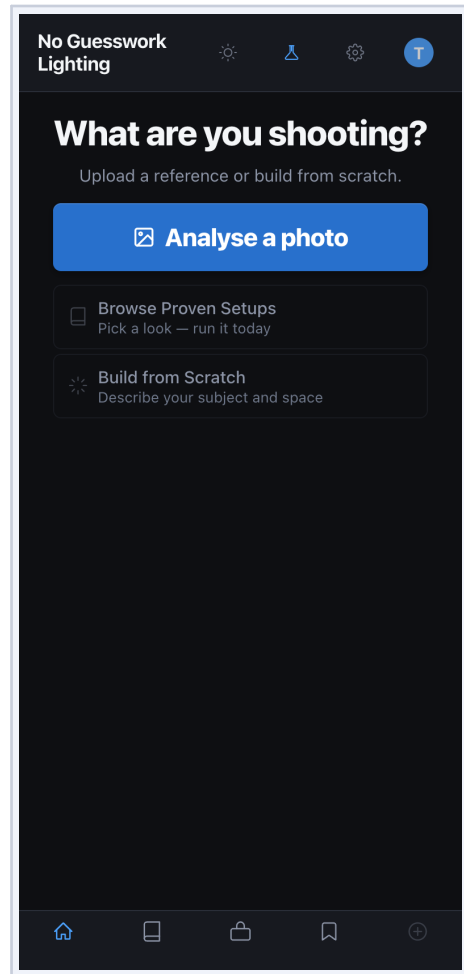


Figure 2 — Home screen: Analyse a photo, Browse Proven Setups, Build from Scratch

- **Analyse a photo** — launches the reference image analysis pipeline
- **Browse Proven Setups** — opens the curated recipe library
- **Build from Scratch** — launches the three-step Setup Wizard
- Recent setups appear as tappable cards below the entry points

■ Note

The header icons give quick access to: Shoot Mode (sun), Lab (flask, admin only), Settings (gear), and your account avatar.

Setup Wizard

The wizard collects three inputs used to select and score lighting systems.

Step	Input	Options	Effect on Results
1	Subject	Portrait, Product, Group, Tabletop, Other	Drives modifier and pattern weighting
2	Environment	Studio, Window, Natural, Mixed, Outdoor	Filters out incompatible systems
3	Mood	Clean, Dramatic, Commercial, Editorial, Beauty, Athletic	Weights scoring toward mood-aligned patterns

After the third step tap **Find My Setup**. The matching engine runs and navigates to the Results screen.

Results Screen

Results are presented as scrollable, collapsible cards. Tap any card header to expand or collapse it.

Card	Contents
Best Match	System name, pattern, reliability score, confidence band
Setup	Per-light cards: modifier, angle, distance ft/m, power stop
Diagram	Top-down floor plan with all lights to scale
Camera Settings	Aperture, ISO, shutter speed, white balance
How to Test	In-camera and on-set verification checklist
Quick Fixes	Symptom → correction troubleshooting guide
Other Setups	Up to 3 alternatives with comparison confidence scores
Space Check	Minimum room dimensions for this setup to work correctly
Skin Tone	Modifier adjustments specific to subject skin tone

Reference Image Analysis

Upload any photograph to identify its lighting. The full 35-pass vision pipeline runs server-side and returns:

- Shadow pattern classification (loop, Rembrandt, clamshell, split, butterfly...)
- Light count estimation and role assignment (key, fill, rim, kicker)
- Modifier inference from catchlight shape and topology
- Recreation guide: modifier type, placement angle, fill strategy

- Confidence score and supporting signal breakdown

■ Note

Well-exposed portraits with visible face shadows and at least one catchlight in the eyes produce the most accurate results. Dark, blurry, or heavily processed images reduce accuracy.

My Kit

My Kit stores your gear inventory so NGW can tailor recommendations to equipment you actually own. The matching engine weights results toward systems buildable from your kit.

- Tap **Add Item** to add a light, modifier, camera, or accessory
- Each item: category, brand/model, quantity, optional notes
- Gear not in your kit shows a "you'll need" indicator on results
- Update your kit when you rent or borrow equipment for a shoot

Shoot Mode

Shoot Mode activates real-time on-set coaching. Launch from the home screen header (sun icon) or from any loaded setup.

1. **Pre-Shoot Checklist** — Equipment verification before you begin
2. **Live Guidance** — Shot-by-shot coaching cards as you work
3. **Problem Solver** — Tap a symptom for instant corrective suggestions
4. **Post-Shoot Review** — Rate the result and save notes for next time

■ Note

Shoot Mode requires a Pro subscription. A 7-day free trial activates on first use.

Room Planner

Enter your studio or shooting space dimensions so that diagram positions display as real-world distances. Access via the More tab → Room Planner.

- Draw walls by dragging on the canvas
- Mark doors, windows, and fixed furniture as obstacles
- Mark the subject position — all distances calculate from this point
- Room plan is saved per-device and applied to all future setups

Settings

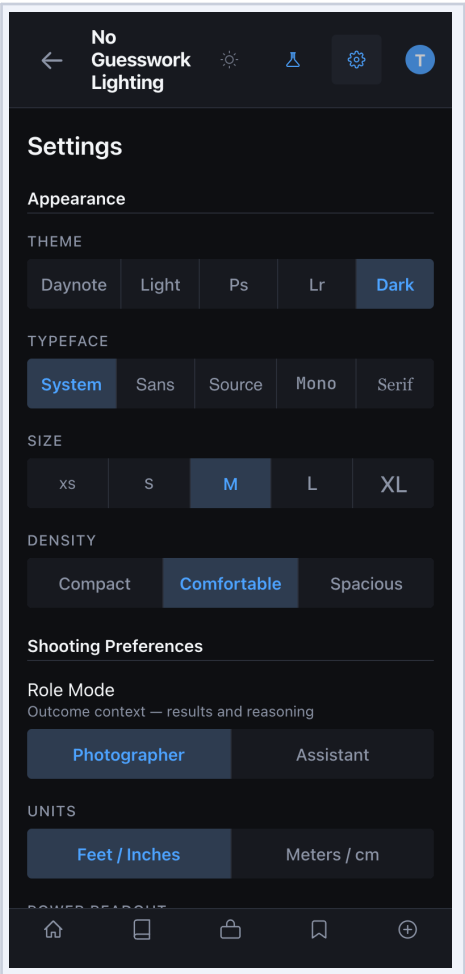


Figure 3 — Settings screen: units, VLM analysis, theme, and account management

Setting	Default	Description
Units	Imperial	Switch to Metric (m/cm) for all diagram distances
VLM Analysis	On	Use AI vision for deeper reference image analysis
Dark Mode	System	Force dark or light theme override
Notifications	Off	Push alerts for saved setup reminders
Account	—	Sign in, upgrade plan, or manage subscription
Lab Access	Admin	Visible only to admin accounts — opens NGW Lab

Recipes (Proven Setups)

The Recipes screen contains a library of pre-built lighting setups organized by subject type and visual mood. Each recipe includes the full setup card stack identical to a Wizard-generated result — modifier, position, diagram, camera settings, and quick fixes.

- Browse by Subject (Portrait, Product, Athletic, Editorial...)
- Filter by Mood (Clean, Dramatic, Commercial...)
- Tap any recipe to load the full results view
- Tap **Save** to add it to My Setups for on-set recall

Saved Setups

Saved setups are stored per-account and sync across devices when signed in. Each saved entry stores the full result payload — all cards, the diagram, and camera settings — so you can reload any previous shoot instantly.

- Tap any saved setup to reload the full results view
- Swipe left (mobile) or tap the trash icon to delete
- Use the search bar to find setups by name or pattern
- Saves are automatic from the Results screen share button → Save Setup

PART III

NGW Lab Manual

Internal development environment for administrators

NGW Lab Manual

The NGW Lab is an internal development environment restricted to admin accounts. It provides tools for testing the analysis pipeline, curating benchmark data, proposing engine improvements, and reviewing the Reference Dataset.

Access & Authentication

The Lab tab and Lab entry on the home screen are visible only to accounts whose email appears in the **ADMIN_EMAILS** environment variable. Admin accounts automatically receive enterprise-level feature access.

1. Sign in with an admin account
2. Tap the **flask icon** in the header, or the NGW Lab entry on the home screen
3. The Lab screen loads with four tabs across the top

■ Important

If "Sign in required" appears you are not authenticated. If the flask icon never appears, your email is not in ADMIN_EMAILS.

Lab Navigation — Four Tabs

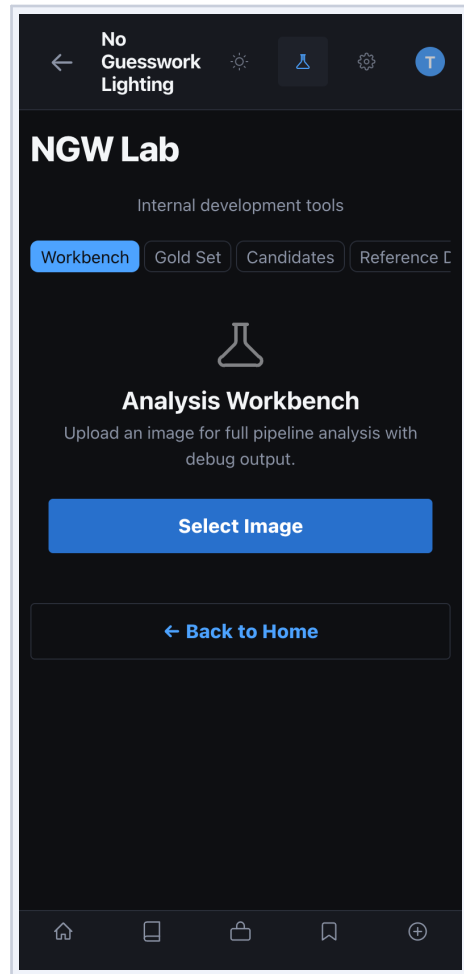


Figure 4 — NGW Lab screen showing the four tabs: Workbench, Gold Set, Candidates, Reference Dataset

Tab	Purpose
Workbench	Run images through the full pipeline; inspect every output layer
Gold Set	Manage the benchmark truth dataset
Candidates	Track proposed engine rule changes draft → accepted
Reference Dataset	Browse, approve, reject, and reprocess the reference library

Workbench

The Workbench is the primary analysis sandbox. It runs the complete NGW pipeline and exposes every intermediate result.

1. Tap **Select Image** and pick a photo (JPEG, PNG, WebP, HEIC, TIFF — max 10 MB)

2. Optionally tick **Debug Overlay** to annotate the image with detected shadows, highlights, catchlights, surface classes, and light roles
3. Tap **Analyze** — the full pipeline runs, typically 5–20 seconds
4. Results appear across four sub-tabs (see below)

Workbench Sub-Tabs

Sub-Tab	Contents
Formatted	Human-readable cards: Description, Narrative, Lighting (family, quality, direction, shadow pattern, fill/rim, light count), Recreation Setup (modifier, placement, fill strategy)
VLM vs CV	Side-by-side VLM-extracted signals vs computer vision signals. Each row has an Accept VLM button to override the CV value.
Raw JSON	Full API response — every signal, candidate, score, and debug field as pretty-printed JSON
Debug Overlay	Annotated image with detected regions drawn on top (only available when Debug Overlay was checked before analysis)

Post-Analysis Actions

- **Save to Gold Set** — Pre-fills a Gold Set entry with this image and analysis; turns green ("✓ Commit to Gold Set") if VLM overrides were accepted
- **Propose Rule** — Pre-fills a Candidate entry with the detected lighting family
- **New Image** — Clears everything and returns to the upload state

■ Note

The VLM vs CV tab is disabled if no VLM API key is configured (OPENAI_API_KEY or ANTHROPIC_API_KEY). The CV pipeline still runs fully.

Gold Set

The Gold Set is the benchmark truth dataset. Each entry pairs a known image with verified expected analysis output. It drives automated accuracy evaluation via python3 scripts/run_benchmarks.py.

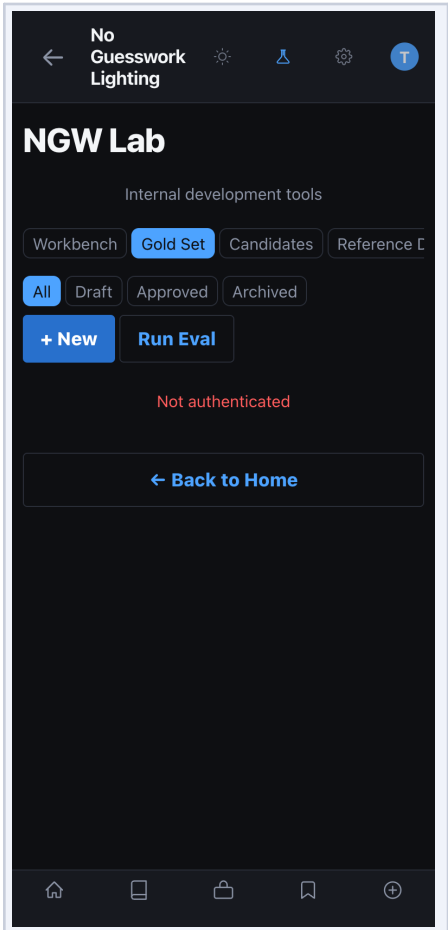


Figure 5a — Gold Set listing with status badges

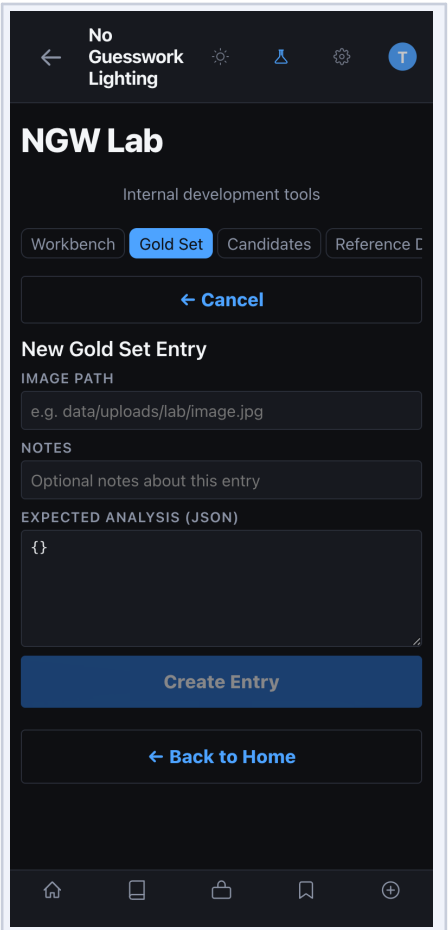


Figure 5b — New Gold Set entry form

Gold Set Status Flow

Status	Meaning	Included in Benchmarks
Draft	Being written or under review	No
Approved	Confirmed correct ground truth	Yes
Archived	Superseded — kept for history	No

■ Important

Archive rather than delete confirmed entries. Deletion is permanent — archiving preserves history.

Candidates

Candidates track proposed engine rule changes. When the Workbench reveals a classification gap, a signal contradiction, or a new pattern worth adding, create a Candidate to document it.

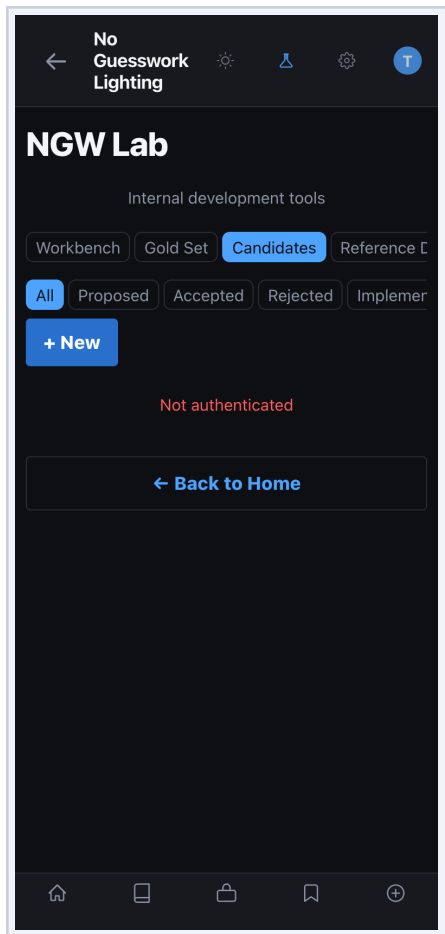


Figure 6a — Candidates listing with status

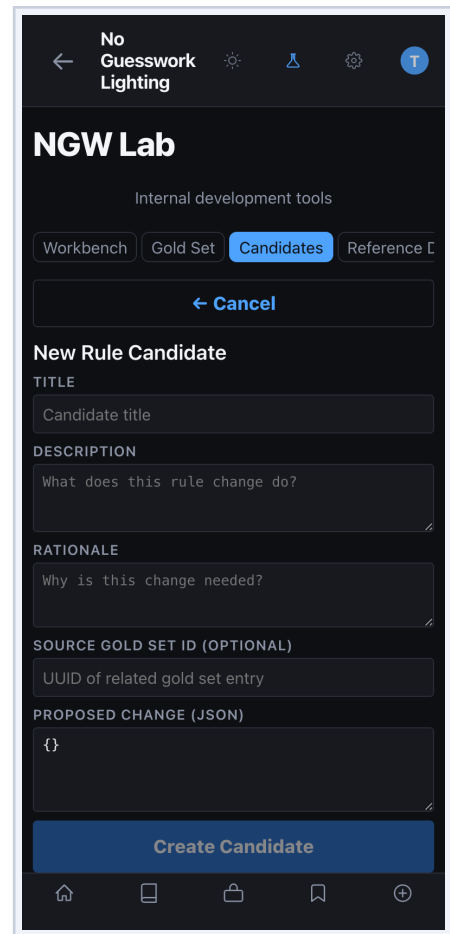


Figure 6b — New candidate form (title, description, rationale)

Status	Meaning
draft	Being written
review	Ready for evaluation against benchmarks
accepted	Merged into engine rules
rejected	Discarded with documented reason

Reference Dataset

A curated library of reference photos with full pipeline analysis attached. Used to build and validate the pattern match library.

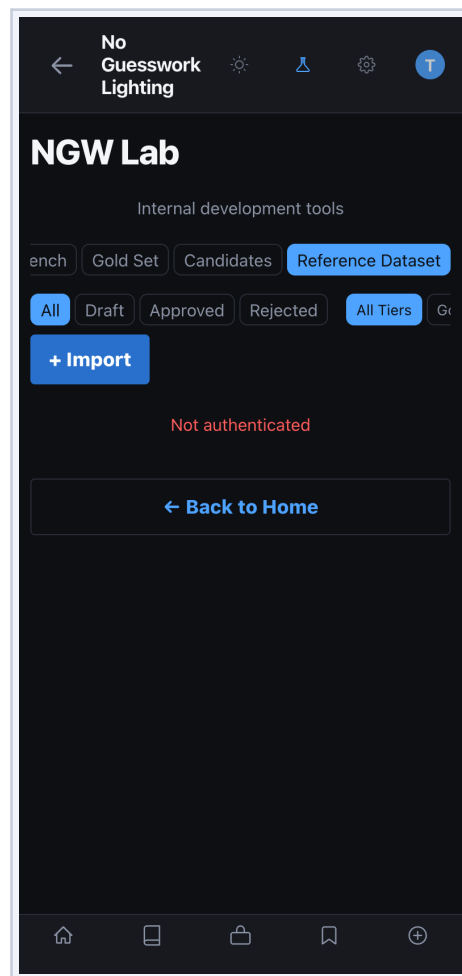


Figure 7 — Reference Dataset grid with pending / approved / rejected status badges

- Browse as a thumbnail grid — tap to open the detail view
- Navigate entries with ← Prev / Next → arrows or the position counter
- Detail view shows: full image, Reference Analysis, Pipeline Signals, VLM Reconstruction
- **Approve** — marks entry as valid reference (eligible for benchmarks)
- **Reject** — marks as unsuitable (filtered, not deleted)
- **Reprocess** — re-runs full pipeline when the engine has been updated

■ Note

New images are ingested via POST /api/lab/reference/ingest — not through the grid UI. After ingestion they appear as "pending".

Lab Settings & Dev Tools

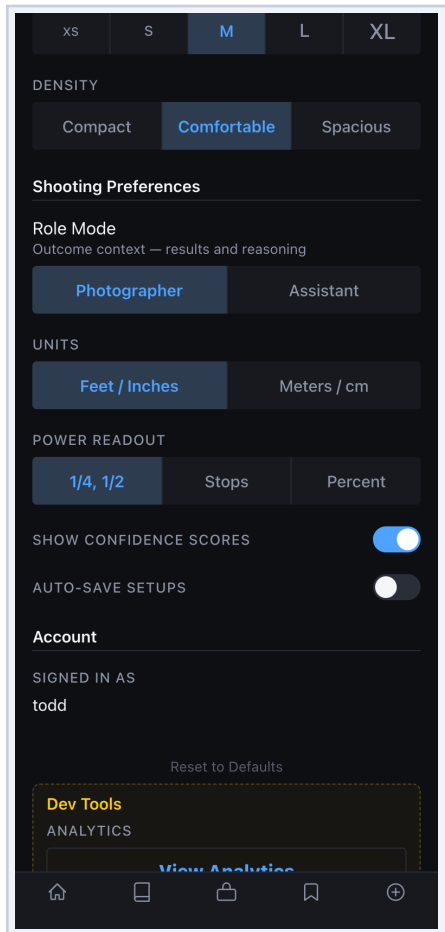


Figure 8a — Lab settings panel

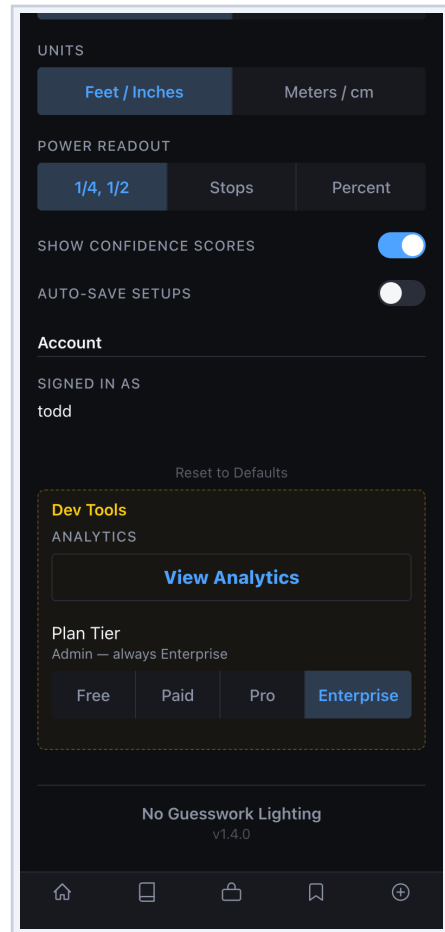


Figure 8b — Developer tools panel

The Settings screen includes two admin-only panels:

- **Lab Settings** — Toggle VLM provider, override model, set debug verbosity
- **Dev Tools** — Inspect active engine constants, trigger cache flush, view server version and benchmark timestamp

Common Lab Workflows

Testing a New Image End-to-End

1. Open **Workbench** → Select Image → Analyze
2. Review **Formatted** — check lighting family, modifier, light count
3. Switch to **VLM vs CV** — accept VLM overrides where the VLM is more accurate
4. Switch to **Raw JSON** — check `bestMatch.reliabilityScore` and the `pattern_candidates` array
5. If results are correct: tap **Save to Gold Set**
6. If the engine misclassified: tap **Propose Rule** with evidence attached

Investigating a SOFT_PASS Benchmark

1. Find the image path in benchmark output
2. Load it in Workbench with **Debug Overlay** checked
3. Inspect the overlay — look for weak catchlights, pose-induced shadows, ambiguous face geometry
4. Check VLM vs CV — identify signal conflicts and which source is correct
5. If a fix is clear: Propose Rule with the relevant signal data cited

Adding a Reference Dataset Image

1. Call POST /api/lab/reference/ingest with the image path
2. Image appears in the Reference Dataset grid as "pending"
3. Open it in the detail view — review analysis, signals, and VLM reconstruction
4. If the analysis is correct: **Approve**
5. If the image is unsuitable (bad framing, ambiguous, duplicate): **Reject**
6. If signals are stale after an engine update: **Reprocess**

Environment Variables

Variable	Required For	Notes
OPENAI_API_KEY	VLM — OpenAI	gpt-4.1 default
ANTHROPIC_API_KEY	VLM — Anthropic	claude-sonnet-4-20250514 default
VLM_PROVIDER	Override	openai / anthropic / none
VLM_MODEL	Override	Full model name string
ADMIN_EMAILS	Lab access	Comma-separated email list

■ Note

VLM is entirely optional. Without it the full CV pipeline still runs — the VLM vs CV tab is disabled but all other Workbench features work normally.

PART IV

Engine Reference

Architecture, pipeline stages, and taxonomy

Engine Reference

This section is the authoritative reference for the NGW analysis engine — all pattern names, signal definitions, classifier precedence, and taxonomy rules. No new categorical values may be introduced without updating engine/enums.py.

Pipeline Architecture — 7 Stages

Stage	Module	Description
1 — Pre-processing	vision_pipeline.py	Resize, color-space conversion, EXIF strip
2 — Region Attribution	vision_pipeline.py	Face detection (MediaPipe), subject bounding box
3 — Signal Extraction	vision_passes.py	35+ signal passes: shadow, highlight, catchlight, geometry
4 — Cue Assembly	cue_extraction.py	Build VisualCueReport from pass outputs
5 — Pattern Matching	pattern_matcher.py	Score each candidate against extracted cues
6 — VLM Reconciliation	vlw_reconciliation.py	Merge VLM signals with CV signals (if configured)
7 — Orchestration	orchestrator.py	Select authoritative pattern, build AnalysisResult

Pattern Classifiers — Priority Order

Four classifiers run in order. Higher-priority classifiers override lower ones when their confidence exceeds the threshold.

Priority	Classifier	Source	Min Confidence
1 (highest)	Shadow Pattern Classifier	cue_extraction.py	≥ 0.65
2	Catchlight Topology Classifier	vision_passes.py	≥ 0.60
3	Geometry & Ratio Classifier	pattern_matcher.py	≥ 0.55
4 (lowest)	VLM Style Classifier	vlw_reconciliation.py	≥ 0.50

Lighting Patterns — 23 Canonical Values

Pattern	Category	Key Characteristic
clamshell	Beauty	Under-chin fill — symmetrical butterfly shadow

Pattern	Category	Key Characteristic
loop	Classic	Small nose-side loop shadow at 30–45°
rembrandt	Classic	Triangular highlight on shadow-side cheek
split	Dramatic	90° key — half face lit, half in shadow
butterfly	Beauty	Direct frontal — shadow below nose only
broad	Fashion	Key on camera-near side, wide coverage
short	Fashion	Key on camera-far side, narrow lit area
triangle	Classic	Hurley triangle — forward-facing, triangular shadow
flat_fashion	Fashion	Ultra-soft frontal, minimal shadow depth
high_key_beauty	Beauty	Bright near-white background, low contrast
low_key	Dramatic	Deep shadow, high contrast, dark background
editorial_rim_key	Editorial	Hard rim as primary with minimal fill
corporate_soft_key	Commercial	45° soft key, clean shadow, neutral background
window_soft_side	Natural	Window as key, fill from bounce or opposite wall
window_negative_fill	Natural	Window key + negative fill panel opposite
ring_light	Beauty	On-axis ring — characteristic ring catchlight
strip_dramatic	Dramatic	Narrow strip for edge/separation light
bare_bulb_editorial	Editorial	Unmodified strobe, deep hard shadows
tabletop_soft_product	Commercial	Overhead + side soft for even product coverage
bottle_backlight	Commercial	Translucent product backlit through diffusion
athletic_rim_sculpt	Athletic	Dual rim + fill — contour sculpting for athletes
soft_editorial_key	Editorial	Large source, low contrast, directional
beauty_dish_clean	Beauty	Beauty dish at 45° — defined shadow, specular pop

Signal Types — 17 Pass Outputs

Signal	Pass	Description
shadow_edge_hardness	shadow	Hard vs soft edge gradient
primary_shadow_direction	shadow	Clock-face direction of key shadow
vertical_light_angle	shadow	Estimated elevation of key light
catchlight_position	catchlight	Clock-face position in iris
catchlight_shape	catchlight	Circle, rectangle, octagon, or ring
catchlight_topology	catchlight	Single, dual, triple, or ring count
highlight_axis_map	highlight	Specular axis direction across face
highlight_symmetry	highlight	Bilateral symmetry ratio
light_structure	geometry	Full light count + role assignment
contrast_ratio	tonal	Lit-to-shadow luminance ratio
background_illumination	background	Brightness + gradient pattern
subject_bg_separation	background	Subject-vs-background separation score
multi_shadow_detection	shadow	Count of distinct shadow patterns
tonal_processing	tonal	B&W; / graded / natural / processed flag
pose_shadow_interference	shadow	Hands or hair occluding face shadows
env_shadow_continuity	environment	Natural light, foliage, window gradient
reflection_architecture	specular	Specular shape on reflective surfaces

VLM Safety Constraints

- VLM may override CV signals only when CV confidence < 0.50
- VLM is ignored for face-geometric signals when a face is detected
- VLM style classification is Priority 4 — cannot override confident CV results
- Rate-limit retry: 2 s → 5 s → 15 s → pipeline continues without VLM on 4th failure

Benchmark Verdicts

Verdict	Meaning
PASS	Pattern matches expected AND reliability score meets threshold
SOFT_PASS	Pattern correct but reliability or signal strength below target
FAIL	Pattern mismatch or critical signal error

- Full run: `python3 scripts/run_benchmarks.py`
- CI run: `bash scripts/ci_benchmark.sh`
- Nightly: `python3 scripts/nightly_benchmark.py`

■ Note

Current scores: 22 PASS · 10 SOFT_PASS · 0 FAIL. All SOFT_PASS cases are perception-limited — the classification logic is correct but signal extraction is ambiguous.

PART V

Appendix

Glossary, reference tables, and operations checklist

Appendix

Glossary

Catchlight	A specular reflection of a light source visible in the subject's eye(s). Shape, position, and count are key signals for modifier and pattern inference.
Clamshell	A lighting pattern using a key light above and a fill light (or reflector) below the subject, creating a symmetrical butterfly shadow.
Gold Set	The curated benchmark dataset of images paired with verified expected analysis outputs. Engine accuracy is measured against it.
Key Light	The primary, dominant light source. Sets the direction, quality, and contrast of the lighting.
Modifier	An attachment that changes the quality or shape of a light source. Examples: softbox, umbrella, beauty dish, grid, bare bulb.
Pattern	A named, reproducible lighting arrangement. NGW recognizes 23 canonical patterns.
Reliability Score	A 0–1 confidence value indicating how strongly the detected signals support the authoritative pattern classification.
Rembrandt	A classic portrait pattern: triangular highlight on the shadow-side cheek, key light at approximately 45° elevation.
SOFT_PASS	A benchmark verdict: the correct pattern was identified but reliability or signal strength is below the target threshold.
VLM	Vision Language Model — an AI model that interprets images. Used as a supplementary signal alongside computer vision.
VisualCueReport	The primary data structure from the cue extraction stage. Contains all extracted signals used by the pattern matcher.

Operations Checklist

Use when deploying or restarting the NGW server:

1. Verify `.env` has all required variables (ADMIN_EMAILS, SECRET_KEY, etc.)
2. Run `python3 -m pytest tests/ -q` — all tests must pass
3. Run `python3 scripts/run_benchmarks.py` — verify 0 FAIL verdicts
4. Confirm `data/systems/canonical/` has 19 YAML files
5. Start: `uvicorn main:app --host 0.0.0.0 --port 8000`
6. Health check: `GET /api/health` → `{"status": "ok"}`

7. Test Lab access with an admin account

Environment Variables Reference

Variable	Required	Default	Description
OPENAI_API_KEY	No	—	Enables OpenAI VLM analysis
ANTHROPIC_API_KEY	No	—	Enables Anthropic VLM analysis
VLM_PROVIDER	No	auto	openai / anthropic / none
VLM_MODEL	No	auto	Override default model name
ADMIN_EMAILS	Yes	—	Comma-separated admin emails
DATABASE_URL	No	SQLite	PostgreSQL connection string
SECRET_KEY	Yes	—	JWT signing secret
CORS_ORIGINS	No	*	Allowed CORS origins